F12 4th CAR

Lecture 23: Section 3.4
Exponential and Logarithmic Equations

One-to-one properties

Inverse properties

Solving exponential equations

Solving logarithmic equations

Find inverse of exponential function

Find inverse of logarithmic function

(4) $\log x = -2$

$10^{\log x} = 10^{-2}$

$x = 10^{-2}$ or $\log_{10} x = -2$

$\boxed{x = \frac{1}{100}}$

check domain

$10^{-2} = x$

$\frac{1}{100} = x$

Lecture 23: Section 3.4
Exponential and Logarithmic Equations

Two basic strategies for solving exponential or logarithmic equations:

1. **One-to-One Properties**

   $a^x = a^y$ if and only if $x = y$

   $\log_a x = \log_a y$ if and only if $x = y$

2. **Inverse Properties**

   $a^{\log_a x} = x$, for $x > 0$

   $\log_a a^x = x$, for $(-\infty, \infty)$

<u>ex.</u> Solve for x:

1) $2^x = 64$ $\quad 2^x = 2^6$ $\quad x = 6$

2) $\ln x - 2\ln 2 = 0$ $\quad \ln x = 2\ln 2$ or $\ln \frac{x}{4} = 0$

   $\ln x = \ln 2^2$ $\quad e^0 = \frac{x}{4}$

   $\ln x = \ln 4$ $\quad 1 = \frac{x}{4}$

   $x = 4$ check domain $\quad x = 4$

3) $e^x = 5$ $\quad 3e^x = 5$

   $\ln e^x = \ln 5$

   $x \underbrace{\ln e}_{=1} = \ln 5$

   $\boxed{x = \ln 5}$

4) $\log x = -2$

Solving Exponential Equations

1. Isolate the exponential expression on one side of the equation.

2. Take the logarithm of each side, then use the Properties of Logarithms to bring down the exponent.

3. Solve for the variable.

ex. Solve for x:

1) $3e^{2-x} = 5$

$e^{2-x} = \frac{5}{3}$

$\ln e^{2-x} = \ln \frac{5}{3}$

$(2-x)\underbrace{\ln e}_{=1} = \ln \frac{5}{3}$

$2-x = \ln \frac{5}{3}$

$\boxed{2 - \ln \frac{5}{3} = x}$

2) $5^{-\frac{x}{2}} = 2$

$\log_5 5^{-\frac{x}{2}} = \log_5 2$

$-\frac{x}{2} \underbrace{\log_5 5}_{=1} = \log_5 2$

$-2\left[-\frac{x}{2} = \log_5 2\right]$

$x = -2\log_5 2$

$\boxed{x = -2\frac{\ln 2}{\ln 5}}$ now use calc.

or $\ln 5^{-\frac{x}{2}} = \ln 2$

$-\frac{x}{2}\ln 5 = \ln 2$

$-2\left(-\frac{x}{2} = \frac{\ln 2}{\ln 5}\right)$

$\boxed{x = -2\frac{\ln 2}{\ln 5}}$

ex. Solve for x: $5^{x-1} = 3^{2-x}$

$\ln 5^{x-1} = \ln 3^{2-x}$

$(x-1)\ln 5 = (2-x)\ln 3$

$x\ln 5 - \cancel{\ln 5} = 2\ln 3 - x\ln 3$

$+\cancel{\ln 5} \qquad +\ln 5$

$x\ln 5 + x\ln 3 = 2\ln 3 + \ln 5$

$x(\ln 5 + \ln 3) = 2\ln 3 + \ln 5$

$x = \dfrac{2\ln 3 + \ln 5}{\ln 5 + \ln 3}$

$x = \dfrac{\ln 9 + \ln 5}{\ln 5 + \ln 3} =$

$x = \dfrac{\ln(9\cdot 5)}{\ln(5\cdot 3)} = \boxed{\dfrac{\ln 45}{\ln 15}}$

$(u)^2 = (e^x)^2$

Quadratic type

ex. Solve for x: $e^{2x} - e^x - 6 = 0$

let $u = e^x$

then $u^2 = e^{2x}$

$u^2 - u - 6 = 0$

$(u-3)(u+2) = 0$

$u = 3 \qquad u = -2$ sub back

$e^x = 3 \qquad e^x = -2$ no solution

$\ln e^x = \ln 3 \qquad \ln e^x = \ln(-2)$

not in domain of ln

$x\ln e = \ln 3$

$x = \ln 3$

Solving Logarithmic Equations

1. Isolate the logarithmic term on one side of the equation; you may need to first combine the logarithmic terms.

2. Write the equation in exponential form (or raise the base to each side of the equation).

3. Solve for the variable.

other way

$2^{\log_2(25-x)} = 2^3$

ex. Solve for x:

1) $1 + \log_2(25 - x) = 4$

$\log_2(25-x) = 3$ $\boxed{x = 17}$ check domain

$2^3 = 25 - x$

$8 = 25 - x$

$2^{\log_2(25-x)} = 2^3$

$25 - x = 8$

$x = 17$

2) $4 + 3\log(2x) = 16$

$3\log(2x) = 12$

$\log(2x) = 4$

$10^4 = 2x$

$10{,}000 = 2x$

$x = 5000$ check domain

can do the other way

ex. Solve for x: $\log_4 x - \log_4(x-1) = \frac{1}{2}$

$\log_4\left(\frac{x}{x-1}\right) = \frac{1}{2}$

$4^{1/2} = \frac{x}{x-1}$

$\frac{2}{1} = \frac{x}{x-1}$

$2(x-1) = x$

$2x - 2 = x$

$\boxed{x = 2}$ ch domain

or $4^{\log_4\left(\frac{x}{x-1}\right)} = 4^{1/2} \rightarrow \frac{x}{x-1} = 2$

ex. Solve for x: $\log(x+2) + \log(x-1) = 1$

$\log(x+2)(x-1) = 1$

$\log(x^2+x-2) = 1$

$10^1 = x^2+x-2$

$x^2+x-12=0$

$(x+4)(x-3)=0$

$\cancel{x=-4}$ $\boxed{x=3}$ check D

$-4+2 \not> 0$

NOTE: Check your answers when solving logarithmic equations.

$\ln_e 1 = ?$

$e^? = 1$

ex. Find the domain of $f(x) = \ln(e^{x+2} - 1)$. >0

$e^{x+2} - 1 > 0$

$e^{x+2} > 1$

$\ln e^{x+2} > \ln 1$

$(x+2)\underbrace{\ln e}_{=1} > \underbrace{\ln 1}_{=0}$

$x+2 > 0$

$x > -2$

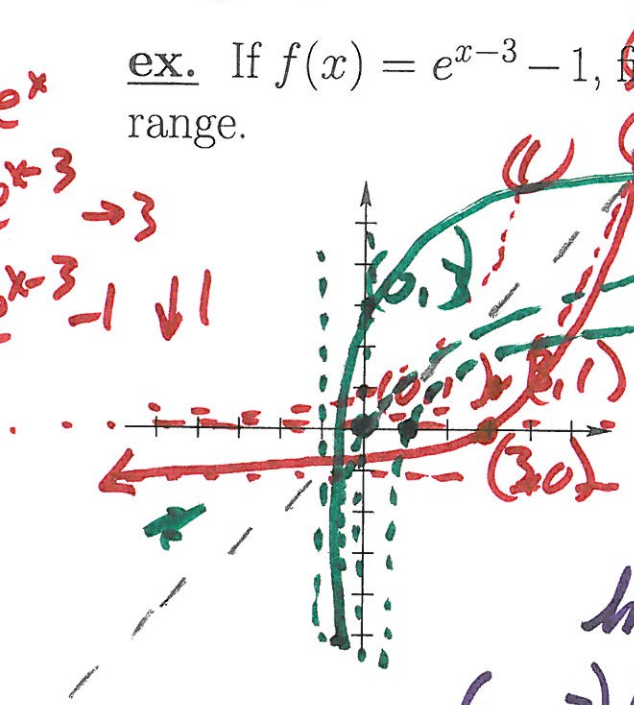
$\boxed{(-2, \infty)}$

ex. If $f(x) = e^{x-3} - 1$, find f^{-1} and its domain and range.

① $y = e^x$

② $y = e^{x-3}$ $\rightarrow 3$

③ $y = e^{x-3} - 1$ $\downarrow 1$



$y = e^{x-3} - 1$

$x = e^{y-3} - 1$

Solve for y.

$e^{y-3} = x + 1$

$\ln e^{y-3} = \ln(x+1)$

$(y-3)\ln e = \ln(x+1)$

$y - 3 = \ln(x+1)$

$y = \ln(x+1) + 3$

① $y = \ln x$

② $\ln(x+1)$

$\leftarrow 1$

③ $y = \ln(x+1) + 3$

$D_f = (-\infty, \infty) = R_{f^{-1}}$

$R_f = (-1, \infty) = D_{f^{-1}}$

$D_{f^{-1}}$: $x + 1 > 0$

$x > -1$

$\boxed{f^{-1}(x) = \ln(x+1) + 3}$

① $y = \log_3 x$

② $y = \log_3(x-4) \rightarrow 4$

③ $y = \log(x-4)+2 \uparrow 2$

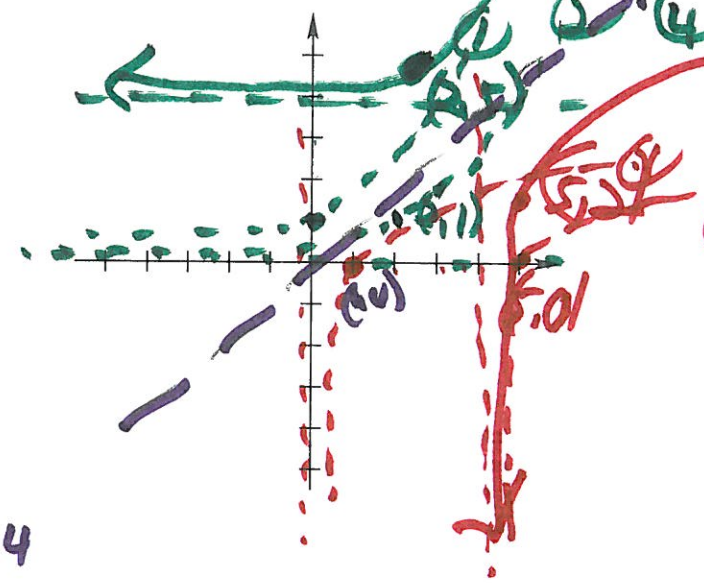
① $y = \log_3(x-4)+2$

Solve for y.

② $x = \log_3(y-4)+2$

③ $\log_3(y-4) = x-2$

ex. Find the inverse of $f(x) = \log_3(x-4)+2$. Find the domain and range of f and its inverse.



$3^{x-2} = y-4$

② $3^{x-2}+4 = y$

$y = 3^{x-2}+4$

$f^{-1}(x) = 3^{x-2}+4$

$x-4>0$

$x>4$

$D_f = (4,\infty) = R_{f^{-1}}$

$R_f = (-\infty,\infty) = D_{f^{-1}}$

① $y = 3^x$

② $y = 3^{x-2} \rightarrow 2$

③ $y = 3^{x-2}+4 \uparrow 4$

$$A = P\left(1+\frac{r}{n}\right)^{nt}$$

P = initial amt = \$5000

$r = 0.09$

Doubled amount is \$10,000 = A

Applications

ex. A sum of \$5000 is invested at an interest rate of 9% per year. Find the time required for the money to double if the interest is compounded according to the following methods.

1) Semiannual $n=2$, solve for t

$$10{,}000 = 5000\left(1+\frac{.09}{2}\right)^{2t}$$

$$2 = (1+.045)^{2t}$$

$$2 = (1.045)^{2t}$$

$$\ln 2 = \ln(1.045)^{2t}$$

$$\ln 2 = 2t\ln(1.045)$$

$$\boxed{t = \frac{\ln 2}{2\ln(1.045)} \approx 7.9 \text{ years}}$$

used a calculator.

2) Continuous $A = Pe^{rt}$ solve for t.

$$10{,}000 = 5000e^{.09t}$$

$$2 = e^{.09t}$$

$$\ln 2 = \ln e^{.09t}$$

$$\ln 2 = .09t\,(\underbrace{\ln e}_{=1})$$

$$\ln 2 = .09t$$

$$t = \frac{\ln 2}{.09} \approx 7.7 \text{ years}$$

used a calculator.

ex. A small lake was stocked with a certain species of fish. The fish population is modeled by the function

$$P = \frac{2}{1 + 4e^{-0.8t}}$$

where P is the number of fish in thousands and t is measured in years since the lake was stocked.

Let $t=0$ 1) How many fish were stocked in the lake initially?

$$P = \frac{2}{1+4e^{-0.8(0)}} = \frac{2}{1+4e^{0}} = \frac{2}{1+4} = \frac{2}{5}$$

so $\frac{2}{5}$ of 1000 is $\frac{2}{\cancel{5}_{1}}\left(\frac{\cancel{1000}^{200}}{1}\right) = \boxed{400 \text{ fish}}$

2) After how many years will the fish population reach 1000 fish? 1000 fish means $P=1$

$$1 = \frac{2}{1+4e^{-0.8t}}$$

$$1 + 4e^{-0.8t} = 2$$

$$4e^{-0.8t} = 1$$

$$e^{-0.8t} = \frac{1}{4}$$

$$\ln e^{-0.8t} = \ln \frac{1}{4}$$

$$-0.8t = \ln \frac{1}{4}$$

$$t = \frac{\ln 4^{-1}}{-0.8} = \frac{-\ln 4}{-0.8} = \boxed{\frac{\ln 4}{0.8} \text{ years} \approx 1.7 \text{ years}}$$

used a calculator

Pratice.

Solve for x:

1) $\log x + \log(x+3) = 1$

2) $\log_2 x = \log_4(3x+4)$

3) $3^{x-2} + 5 = 32$

4) $9^x + 3^x - 6 = 0$

Answer. 1) 2 2) 4 3) 5 4) $\log_3 2$